

Mollusks

PHYLUM	Molluska
CLASSES	7
ORDERS	53
FAMILIES	609
SPECIES	About 110,000

Of all invertebrates, mollusks have arguably the widest range of body forms. They include gastropods (slugs and snails), bivalves (such as oysters and clams), cephalopods (octopuses, squids, and cuttlefishes, among others), chitons, and several less familiar forms. All mollusks have one or more

of the following: a horny, toothed ribbon in the mouth (the radula) that is unique to mollusks; a calcium-carbonate shell or other structure covering the upper surface of the body; and a mantle or mantle cavity, typically with a characteristic type of gill. Mollusks are adapted to life on land and in water; there are also many parasitic species.

Movement

Most mollusks use their broad, flat foot to creep over hard surfaces, often on a thin layer of mucus. Some bivalves are fixed to the surfaces on which they live, and all are relatively inactive, but they often have a well-developed foot, which some species use to dig burrows in sediment. The most mobile and fastest-moving of all mollusks are the cephalopods, especially octopuses and squids. Using undulations of their muscular fins or by flapping them, they can make a slow swimming motion but they are also able to move more rapidly by squirting a jet of water from the mantle cavity.

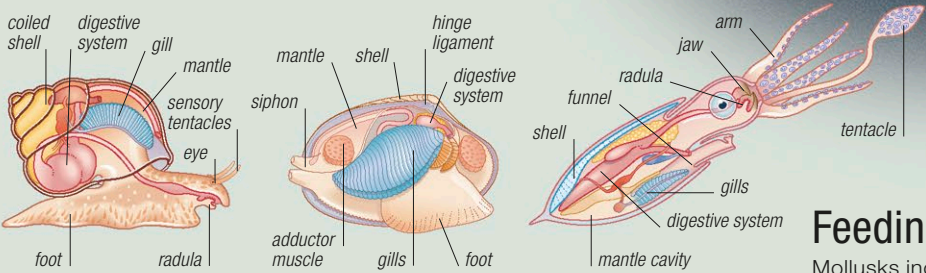
MUCUS TRAILS

To reduce surface friction, slugs and snails secrete mucus from their foot. Slugs (such as this striped slug) have lost their shells, and generally move more quickly than snails.



Anatomy

Most mollusks consist of 3 parts: a head, a soft “body” mass, and a muscular foot, which is formed from the lower surface of the “body” mass and used for locomotion. In some mollusks, the head is well developed and supports sophisticated sense organs, but in others (such as the bivalves), it is effectively absent. The upper surface of the “body” mass is covered by a layer called the mantle, from which the shell is secreted. Retractor muscles can be used to pull the shell over the body for safety. Besides protection, the shell may be used as an organ for burrowing or controlling buoyancy. A fold in the mantle forms a cavity that in aquatic mollusks houses the gills.



GASTROPOD

Gastropod snails have a coiled shell, a large foot, and a distinct head, with eyes and sensory tentacles. Slugs have no shell.

BIVALVE

Bivalves are enclosed within a pair of shells, from which the foot and siphons can be extended. The gills are often large and folded.

CEPHALOPOD

In cephalopods, part of the head and foot are modified to form arms and a funnel into the gill chamber. The shell is small or absent entirely.



BIVALVE SHELL

The 2 halves of a bivalve's shell are joined along one edge by elastic material and closed by contracting the adductor muscles.



GASTROPOD SHELL

In most snails, the shell is secreted unevenly from the mantle, which causes it to coil into a clockwise spiral (occasionally counterclockwise).

Feeding

Mollusks include predators, grazers, and browsers, as well as species that feed on deposited or suspended material. Many bivalves and a few gastropods are filter feeders, using their gills to strain particles as small as bacteria from water. Cephalopods are carnivores. They use their long arms, which are often equipped with suckers, to catch fishes, crustaceans, and other prey. These relatively intelligent animals have been known to stalk prey. Besides their radula, cephalopods have a parrotlike beak, which they use to bite their prey. Most other mollusks feed on food ranging from attached algae, plants, and animals to planktonic jellyfish and even slow-moving fishes. Parasitic species obtain nourishment from their host.

SQUID SCHOOL

All cephalopods are marine. Octopuses tend to be solitary but squids commonly form schools, species of the genus *Loligo* being particularly sociable. Cuttlefishes only congregate in the breeding season.



HUNTING

Some octopuses are capable of subduing large and formidable prey. This one has overpowered a zebra moray eel.



RADULA

The tiny, toothlike denticles on the radula, which may number up to several thousand, are used like a rasp to collect small particles of food.